

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 08

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Normal diagonal mesh closes as the net fills / stretches so small fish cannot escape; square-mesh panels do not close so juveniles escape to grow and breed; TEDs let turtles out; larger mesh sizes generally (2) • Minimum size lets fish reach breeding age before capture (eg UK lobster minimum carapace above breeding size, cod 35 cm); maximum size / V-notching egg-carrying females protects large, highly fecund breeding adults (Hebridean lobsters over 145 mm, Nassau grouper) (2) • NTZs (Lundy, Lamlash Bay, Chagos, Great Barrier Reef) protect breeding populations and habitats; surplus young spill over and increase surrounding catches; damaged communities recover (2) • Closed seasons, ideally in the breeding season, let fish grow and breed; fishermen may oppose because shoaling breeders are easy to catch (2)	8
01.2	<i>1 mark for each point.</i> • Pelagic species form single-species shoals so fishing stops when the quota is reached (1) • In mixed fisheries many species are caught together; once one species' quota is reached it must be discarded (dead) while fishing for others continues (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
02.1	<i>1 mark for each point.</i> • Coordinating global data on biodiversity conservation (Red List) (1) • Increasing understanding of the importance of biodiversity (1) • Deploying nature-based solutions to global challenges in climate, food and sustainable development (1)	3

Question	Marking guidance	Marks
02.2	<i>1 mark for each point.</i> - Decline = $(12\,000 - 2100) / 12\,000 \times 100 = 82.5\%$ (1) - Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km^2) and adults (2100) only meet Endangered (1) - A species is placed in the highest category for which it meets any one criterion, so Critically Endangered (1)	3
02.3	<i>1 mark for each point.</i> - Categorising a species requires a great deal of research (population, range, trends) and resources; about 85 000 of millions of species assessed (1) - Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) (1) - Evolutionarily unique species / EDGE species; endemic species; keystone species; flagship species; species with fragmented populations (1)	3
02.4	<i>1 mark.</i> Iberian lynx: Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years); Fregate Island beetle after rat eradication (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	
03.1	<i>1 mark for the correct answer.</i> $2.7 - 2.45 = 0.25$ billion = 250 million years (1)	1
03.2	<i>1 mark for each point.</i> - For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans (1) - Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere (1)	2
03.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Oxygen in the atmosphere absorbed ultraviolet light (1) - A dynamic equilibrium of reactions between O, O₂ and O₃ (the ozone layer) developed (1) - Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) (1) - Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved; many anaerobic Archaea and bacteria died out (1)	3
03.4	<i>1 mark for each point.</i> - Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) (1) - The lower CO₂ concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years) (1)	2
03.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion (1) - Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas (1) - Regulation of atmospheric composition (eg oxygen / CO₂ by photosynthesis and respiration) (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	
04.1	<i>1 mark for each point.</i> • No major temperature extremes; regular water supply with no pronounced dry season; deep fertile soils built by leaf litter and root binding; strong seasonality (1) • Deciduous canopy present spring to autumn shades the floor during the best growing temperatures (1) • Bluebells, snowdrops, lesser celandine grow and flower early in spring before the trees leaf (1) • Winter food shortage causes animals to hibernate (dormice), store food or migrate (1)	4
04.2	<i>1 mark for each point.</i> • Long time for species to colonise; some woodland plants colonise very slowly so their presence (ancient woodland indicators) shows the wood is old (1) • Natural woodland has trees of all ages with clearings where old trees died, giving varied light and dead wood habitats (1) • A new plantation has a single-age dense canopy with low light and few niches until trees start to die (over 200 years) (1)	3
04.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Clearance for farmland, plantations, urban development, roads, mineral extraction (1) • Fragmentation isolating remaining woods so populations cannot move between them and are vulnerable (1) • Management change: end of coppicing so woods become uniform and dark; replacement by conifer monocultures (1) • Overgrazing by deer preventing regeneration; introduced pathogens (ash dieback) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	
05.1	<i>1 mark for each point.</i> • Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala); increases evaporation (1) • Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase (1) • Evapotranspiration changes: increased in irrigated arid areas, decreased where forests were cleared (1)	3
05.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • Seasonal / local: avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes (2) • Organic: uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species; free-range and Fairtrade also accepted for welfare / social reasons (2)	4

Question	Marking guidance	Marks
05.3	<i>1 mark for each point.</i> • Food shortages after 1945 (dependence on US food aid); farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock (1) • Guaranteed market / price support: government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output (1) • Consequences: loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution; later food surpluses (butter mountains) that could not be sold without harming LEDC producers (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
06.1	<i>1 mark for each point.</i> • As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25) (1) • As otters recovered after 1990 urchins declined and kelp recovered (1) • Otter and kelp trends are positively correlated; urchins negatively correlated with both (1)	3
06.2	<i>1 mark for each point.</i> • Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise (1) • Sea urchins graze kelp so the kelp forest was over-grazed and destroyed (1) • Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade); otter recovery restored the balance (1)	3
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Over-fishing of sandeels reduced puffins (1) • Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased; over-fishing of sardines / herring (competitors) also increased jellyfish (1) • Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction; loss of elephants / large herbivores reduces seed dispersal of trees (1)	2
06.4	<i>1 mark for each point.</i> • They control abiotic features: beaver dams create ponds colonised by aquatic species; elephants create clearings and keep water holes open (1) • Species that depend on these features (light, water, minerals) cannot survive in their absence (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.	
07.1	<i>1 mark for each point.</i> • Energy: low energy-density renewable (solar) / non-renewable fossil fuels (1) • Materials: cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion (1) • Wastes: biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate (1) • Temperature: low / high (Haber process, incineration) (1)	4

Question	Marking guidance	Marks
07.2	<i>1 mark for each point.</i> - Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) (1) - They are re-used in cycles so there is no depletion of raw materials or build-up of wastes (1)	2
07.3	<i>1 mark for each point.</i> - Use often disperses the minerals (eg in products spread over the world / corrosion) (1) - Or produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced (1)	2
07.4	<i>1 mark for each point.</i> - They cannot easily be removed from end-of-life appliances and are not biodegradable (1) - Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	
08.1	<i>1 mark for each point.</i> - Moral belief: protecting the environment is the 'right' thing to do regardless of benefit to people (1) - Enlightened self-interest: protecting the environment is done because it causes fewer difficulties (for people) than not doing so (1)	2
08.2	<i>1 mark for each point.</i> - Total impact = population size x the individual (per capita) impact of each person (1) - Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact (1)	2
08.3	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> - Slash and burn agriculture (1) - Fishing (1) - Discharge of sewage / biodegradable pollutants (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
09.1	<i>1 mark for each point.</i> - Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids (1) - Desired characteristics introduced without linked unwanted ones; faster (1)	2
09.2	<i>1 mark for each point.</i> - Bt: reduces insecticide spraying so fewer non-target deaths, lower cost / energy (1) - Roundup Ready: weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage (1) - Golden rice: provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa (1)	3

Question	Marking guidance	Marks
09.3	1 mark for each point. • Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) (1) • Antibiotic-resistance marker genes could transfer to pathogenic bacteria; possible allergies; gene transfer to gut bacteria (1) • Detection in the human food chain from crops approved only for animal feed (1) • Economic: seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs; herbicide-resistant crops may increase herbicide use (1)	4
09.4	1 mark. • Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • In-situ: species remain in natural conditions with the whole community and inter-species relationships; protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • Limits of in-situ: external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • Ex-situ: captive breeding when in-situ is impossible or insufficient; selection criteria; difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation); techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning); seed banks; successes (Hawaiian goose, Iberian lynx, red kite, large blue) • Limits of ex-situ: expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • Judgement: ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to; both needed, in-situ fundamental	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	In-situ conservation keeps a species in its natural habitat with the whole community it depends on - food species, pollinators, predators and the abiotic conditions it is adapted to. It is delivered through protected areas such as SSSIs, NNRs, SACs, SPAs, Ramsar sites and Marine Conservation Zones, land ownership by conservation bodies, agri-environment schemes and habitat creation and management. Its weaknesses are that external threats such as pollution and climate change cross reserve boundaries, enforcement and cost, conflicts with other land uses, and the fact that many populations are already small and fragmented.	
	Ex-situ captive breeding is used when in-situ protection is impossible or insufficient. It is hard: animals need enough space, the right food, social groupings and breeding triggers, and small founder populations risk inbreeding and hybridisation, which studbooks, artificial insemination and embryo transfer help to manage. It is expensive, works for few species, and captive animals may lose wild behaviour; release succeeds only where suitable habitat exists and with soft release and post-release monitoring.	
	The two are not alternatives. Ex-situ methods are a temporary safeguard that can rescue a species from extinction, as with the Arabian oryx, but they only succeed if there is protected habitat to return to. In-situ conservation is therefore fundamental and ex-situ a support for it; the relative importance depends on how close to extinction the species is.	
	Weaker answer - Level 2 (5 marks)	
	In-situ conservation means protecting species in their natural habitat, for example in nature reserves and national parks. This is good because the species stays in the wild with the other species it needs. Ex-situ conservation means captive breeding in zoos and then releasing the animals back into the wild. This can save species that are nearly extinct but it is expensive and animals may not survive when released. Both methods are important and are often used together.	
	What makes the difference	
	• Level 2 defines both approaches correctly but Level 3 gives specific mechanisms (types of protected area, breeding difficulties, release methods) and named examples.	
	• Level 3 weighs the limitations of each approach explicitly; Level 2 gives one limitation for ex-situ only.	
	• 'Both are important' is not a judgement about <i>relative</i> importance; Level 3 explains why in-situ is fundamental and when ex-situ matters most.	
10.2	<i>1 mark.</i> • Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Sustainability definitions; population and per capita impact; attitudes of rural and urban societies • Negative feedback: cloud cover and albedo, CO₂ and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • Tipping points: rare under negative feedback; large events (Permian methane hydrate); self-sustaining change from human activities • Diversity and resistance: reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides); links to climate change chapter; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25